



Like detergents, which become 'New & Improved' with every passing year, computers too keep getting revamped—with progressively fancier badging. The stuff in washing soap that warrants an upgrade (active oxygen, stain digestors, et al) may sound dubious, but there's no doubt what makes a computer meaner and faster—it's all in the enhancement to the CPU or Central Processing Unit.

The CPU may sound boring, but it's the component of a computer system people refer to most while describing the PC they have (such as a Pentium 4, a Celeron, or an AMD). The CPU can be called the brain of your computer. However, this brain requires a body and limbs to function effectively. Even if you have a fast processor, the rest of the components must be able to handle the output of that processor—or you'll be saddled with a system that crawls rather than zips. Keep this in mind when you decide on the right CPU.

There are a number of CPU manufacturers, including AMD, VIA, Motorola—and the grand-daddy of them all, Intel, which has the largest share of the consumer market. Intel's Pentium 4 CPUs are high-speed processors, ranging from 1.4 GHz to 3.06 GHz. However,

the AMD line of CPUs—which include the economy-focused Duron and the high-performance Athlon—has become increasingly popular, as they cost less and offer comparable performance. AMD has also made great strides in the last 2 years in terms of a valid alternative to the Intel line of processors.

So which is the right CPU for you? Well, if you are buying a new one, and want the fastest computer available for graphics, games, and Internet applications, look no further than an Intel Pentium 4 3.06 GHz. This processor supports the latest performance-enhancing technology from Intel, called HyperThreading.

If you cannot afford a Pentium 4 processor, consider the AMD Athlon XP 3000+. If you wish to upgrade in the future, you might want to buy an Intel P4 2.0 or 2.53 GHz, which will give reasonable power for today's applications.

If you're running on a wafer-thin budget but still desire power, go in for the AMD Athlon 1.7 GHz.

Those who want to spend only the absolute minimum can go in for VIA-based processors, but please bear in mind that you may have to say goodbye to all your high-end games and resource-hungry applications.

Market Overview

The last year or so has been a dramatic one for the processor market. The race intensified even as the industry saw a massive jump in processing power from 2 GHz to 3 GHz. With vendors vying with each other to introduce higher speed processors, prices have taken a beating—and Intel and AMD have been pushing the P4 and the Athlon XP respectively.

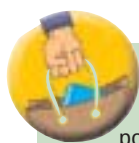
A scant 3 years ago, the Indian processor market did not compare favourably with what was available internationally. But now, it's on par with the western world, and any processor launched internationally is available in India right away. Entry-level processors from all three major players—Intel, AMD and VIA—are available for Rs 3,500 and up. Intel's entry-level workhorse, the Celeron 1.7 GHz, is still going strong with a price range of Rs 3,500 to Rs 3,800.

On the other hand, AMD's entry-level Athlon, the XP 1700+, has a price tag between Rs 3,800 and 4,000. But the majority of users in India still prefer Intel's P4 1.7 and 2.4 GHz, which come for Rs 8,000 to Rs 11,000. And although higher-speed CPUs (such as the P4 3.06 GHz) are available, the price tag is very high—around Rs 25,000 to Rs 28,000.

In the past, we've seen a major price cut from Intel, specifically in the P4 segment—the maximum price drop has happened in the 3.06 GHz, almost 32 per cent. In the next couple of months, this segment could once again observe a major drift in terms of technology and price, with both rivals Intel and AMD planning to shift to 64-bit architecture CPUs with their Itanium2 and Opteron respectively.

The thrust today from vendors as well as resellers is to provide a low-cost PC solution to the masses. VIA, the minnow in the pond, has taken a massive initiative and launched a motherboard that has an onboard processor with all the relevant peripherals. This has also helped AMD gain market share drastically, with market sources saying that in one year it's doubled its sales figure from 7.5 per cent to around 16 per cent.

This segment has been facing some unique problems, especially that of refurbished and repackaged CPUs being sold in the market. Refurbished or used CPUs from a certain area are bought back by resellers and system integrators and shipped to countries that lag behind in terms of technology. There's also a possibility that since a couple of CPUs come in tray packaging for volume sales, they cost less, and are hence bought and repackaged in boxes to get good price margins. Although this phenomenon may not be very prevalent, it does occur at times, especially at times when there is shortage, and you have to learn to be on guard.



Buying tips

- Present-generation CPUs are power-hungry, so ensure that your power supply is rated at 350 W at the least.

- While buying a CPU, either from Intel or AMD, make sure that you buy a genuine one. Genuineness can be confirmed by the hologram seal on the Intel CPU box. In the case of AMD, make sure that the plastic casing is not broken.

- If you plan to do CPU-intensive tasks such as graphics processing, animation and gaming, opt for a CPU that has 256 KB or more of L2 cache. The Pentium 4 Northwood, for example, has 512 KB of L2 cache.

- The new AMD processors sporting the Thoroughbred core name require less core voltage and have lower power dissipation, which makes them expensive. So if power requirement and heat dissipation are not an issue, you could opt for the more affordable Palomino cores.

- Invest in a good heatsink and fan combination for your processor, even if you do not plan to overclock your system.